14.3.1 Classification of lipids:

Based on their structures lipids can be classified as simple lipids, compound lipids and derived lipids. Simple lipids can be further classified into fats, which are esters of long chain fatty acids with glycerol (triglycerides) and waxes which are the esters of fatty acids with long chain monohydric alcohols (Bees wax).

Compound lipids are the esters of simple fatty acid with glycerol which contain additional groups. Based on the groups attached, they are further classified into phospholipids, glycolipids and lipoproteins. Phospholipids contain a phospho-ester linkage while the glycolipids contain a sugar molecule attached. The lipoproteins are complexes of lipid with proteins.

14.3.2 Biological importance of lipids

1. Lipids are the integral component of cell membrane. They are necessary of structural integrity of the cell.
2. The main function of triglycerides in animals is as an energy reserve. They yield more energy than carbohydrates and proteins.
3. They act as protective coating in aquatic organisms.
4. Lipids of connective tissue give protection to internal organs.
5. Lipids help in the absorption and transport of fat soluble vitamins.
6. They are essential for activation of enzymes such as lipases.
7. Lipids act as emulsifier in fat metabolism.

14.4 Vitamins:

Vitamins are small organic compounds that cannot be synthesised by our body but are essential for certain functions. Hence, they must be obtained through diet. The requirements of these compounds are not high, but their deficiency or excess can cause diseases. Each vitamin has a specific function in the living system, mostly as co enzymes. They are not served as energy sources like carbohydrates, lipids, etc.,

The name 'Vitamin' is derived from 'vital amines', referring to the vitamins earlier identified amino compounds. Vitamins are essential for the normal growth and maintenance of our health.

14.4.1 Classification of vitamins

Vitamins are classified into two groups based on their solubility either in water or in fat.

Fat soluble vitamins: These vitamins absorbed best when taken with fatty food and are stored in fatty tissues and livers. These vitamins do not dissolve in water. Hence they are called fat soluble vitamins. Vitamin A, D, E & K are fat-soluble vitamins.

Water soluble vitamins: Vitamins B (B_1, B_2, B_3, B_5, B_6, B_7, B_9 and B_{12}) and C are readily soluble in water. On the contrary to fat soluble vitamins, these can't be stored. The excess vitamins present will be excreted through urine and are not stored in our body. Hence, these two

XII U14-Biomolecules_New.indd 258 08-12-2021 22:52:46